

Garv Arora

Robotics & Embedded Systems Engineer | Autonomous Systems | ADAS

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SUMMARY

Autonomous systems engineer specializing in ROS2 navigation, real-time sensor fusion, and ARM bare-metal firmware — with 2+ years shipping competition-grade robotics software for planetary rover and drone platforms. Placed 12th (IRC 2025) and 16th (IRC 2026) globally among 100+ international teams. Equally at home writing CUDA-accelerated perception pipelines and flashing firmware to microcontrollers. Seeking roles in robotics software, embedded systems, or ADAS.

EDUCATION

Vellore Institute of Technology | Vellore, India

Bachelor of Technology – Computer Science & Engineering (Specialization: IoT)

Gems International School | Gurugram, India

Central Board of Secondary Education (CBSE)

CGPA: 8.37 / 10.0

August 2023 – July 2027

Class 10: 89.3% | Class 12: 76.4%

November 2016 – April 2023

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, SQL, Bash, Verilog, C#, HTML/CSS

Robotics & Autonomous Systems: ROS2, Nav2, MoveIt2, RTAB-Map, Micro-ROS, SLAM, EKF/UKF, PID Control, Point Cloud Processing, mmWave Radar

Perception & ML: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, HuggingFace Transformers

Embedded & Infrastructure: ARM Cortex-M, ESP32, I2C/SPI/UART, FreeRTOS, Docker, Git, GitLab CI/CD, Linux, AWS, PyQt, MongoDB

AI Agents & Backend: LangGraph, FastAPI, Redis, PostgreSQL, Qdrant

WORK EXPERIENCE

SEDS Projects VIT | SEDS India

April 2024 – Present

Research & Development Lead

January 2026 – Present

- Promoted to R&D Lead overseeing 2 competition teams (50+ engineers) — Team Ardra (ISDC) and Team Vyadh (IRC/IRDC); guided 3 sub-team leads through architecture reviews, resolving 10+ pre-competition issues.

Autonomous Systems Developer

April 2024 – January 2026

- Architected a full ROS2 navigation stack (Nav2, RTAB-Map Visual SLAM, RealSense D455), cutting teleoperation time 70%; slashed CV latency ~60% and raised throughput 4x via CUDA, YOLOv8, ArUco/AprilTag, and Linux V4L2 tuning (4 streams).
- Flashed Micro-ROS firmware to ESP32 microcontrollers, fusing IMU and encoder data with an EKF (odometry drift cut ~40%, sub-10 cm positioning); shipped a PyQt5 dashboard with real-time telemetry and FSM-based sequencing.

Samsung R&D Institute | Samsung PRISM Program

May 2026 – Present

Data Science Intern

Bangalore, India

- Selected as 1 of 4 students university-wide for Samsung's PRISM Data Science internship; developed an AI data pipeline using LangGraph and FastAPI.

LG Soft India Pvt. Ltd.

June 2026 – June 2026

Embedded Systems Intern

Greater Noida, India

- Engineered embedded firmware and sensor/microcontroller integration for smart-appliance systems.

Wissen Baum Engineering Solutions

May 2025 – July 2025

Software Automation Intern

Pune, India

- Authored a BDD test suite (Python, Behave, pytest; 40+ scenarios), raising regression coverage 0% to 100% on 3 core product flows.
- Accelerated GitLab CI/CD by parallelizing 3 stages (Playwright + Cypress), trimming run time ~35% (18 min to 12 min); automated QA via pandas/NumPy (10,000+ rows/run).

PROJECTS

mmWave Earthquake Survivor Detection Robot (Patent Filed) | ROS2, Python, Hi-Link LD2410C, ESP32 2026

- Interfaced an LD2410C mmWave radar with ROS2 to detect survivor breathing signatures through 30 cm of rubble; fused radar point-cloud with IMU odometry, flagging coordinates with <15 cm error.
- Designed a confidence-grid mapping algorithm with an explore/confirm/log state machine; Co-Inventor, Patent App. IN202641072249.

Gesture-Controlled 5-DOF Robotic Arm | C++, Arduino, MPU6050, Flex Sensors

2026

- Mapped MPU6050 orientation and 2 flex sensors to 5-DOF servo commands (<50 ms latency); trained classifier on 200+ samples, 94% accuracy across 8 gestures.

HayaiOS – Bare-Metal RTOS | C, ARM Assembly, ARM Cortex-M4

2026

- Implemented a preemptive scheduler (1 ms tick), context switching in <50 assembly instructions, mutex/semaphore primitives, and a full HAL isolating register-level I/O on ARM Cortex-M4.

3D Reconstruction – IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK

2025

- Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth and IMU via complementary filter, reducing tracking failures by ~50% across a 6-DOF trajectory.

CERTIFICATIONS & ACHIEVEMENTS

IRC 2025: 12th globally | **IRC 2026:** 16th globally | **IRDC 2025:** Top 5 Finalist | **Patent Filed 2026:** IN202641072249

AWS Certified: AI, Cloud Practitioner | **OCI:** Gen AI, Data Science | **Amazon ML Summer School:** 1 of 3,000